

Graph-to-Text Generation with Large Language Models: A brief Analysis

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Abstract

This report investigates the use of Large Language Models (LLMs) for graph-to-text (G2T) generation. Using the WebNLG dataset within the GEM benchmark, the study found similar performance across various LLMs, suggesting limitations in task complexity and evaluation metrics like BLEU and BERTScore. The research highlights challenges such as data scarcity in non-English languages and the need for more diverse datasets, advanced graph-aware architectures, and nuanced evaluation methods to improve the generation of accurate and linguistically diverse text from structured data.

1 Introduction

This report analyses the project that evaluates large language models (LLMs) on graph-to-text (G2T) generation tasks using the WebNLG dataset within the GEM benchmark framework (Gardent et al., 2017; Mille et al., 2024). The project addresses the convergence of two major NLP challenges: processing structured graph data (which requires linearization for LLMs) and achieving robust text generation (Ribeiro et al., 2021). The research investigates how effectively advanced LLMs can interpret structured data represented as triplets and verbalize this information into coherent, semantically precise text across different languages—a capability critical for enhancing global information accessibility and making structured knowledge from resources like DBpedia and Wikidata comprehensible to diverse linguistic communities worldwide.

2 Methodology

This project converts structured graph data into natural language using a GNN-enhanced framework. The core of the methodology is to fine-tune a model that directly processes graph structures, allowing it to generate text that is not only fluent but also

faithful to the relational information embedded in the data.

2.1 Dataset and Data Preparation

The study utilizes the English portion of the **WebNLG dataset**, a standard benchmark for graph-to-text generation (Gardent et al., 2017). This dataset contains approximately 13,000 graph-text training pairs derived from DBpedia, covering diverse domains such as Astronaut, University, and Monument.

The graphs in WebNLG are relatively small and self-contained, consisting of 1 to 7 Resource Description Framework triplets. These triplets form distinct structural patterns, categorized as chains (e.g., subject1 -> property1 -> object1/subject2 -> property2 -> object2), sibling relationships (e.g., subject1 -> property1 -> object1 and subject1 -> property2 -> object2), or mixed combinations of both. Unlike methods that rely on linearizing graph structures into a simple sequence of text, our approach is designed to process the graph's topology directly, preserving its relational integrity.

2.2 Model Architecture

The core of our methodology is a dual-encoder architecture, as illustrated in Figure 1. This architecture is inspired by the work done by (Ribeiro et al., 2020), designed to integrate contextual information from both a natural language query and a structured knowledge graph.

The process begins when a query is handled by two parallel paths:

1. **Text Encoder:** The query is fed into a Transformer-based encoder (LLAMA2-7B) (Vaswani et al., 2023; Touvron et al., 2023), which generates a contextual embedding that captures the semantic meaning of the input text.

2. **Graph Encoder:** Simultaneously, a relevant subgraph is retrieved from the knowledge base. This subgraph is then processed by a Graph Neural Network (GNN) Encoder (Scarselli et al., 2009). This component is crucial as it creates a graph-aware embedding that represents the nodes and their interconnections, preserving the structural and relational information of the data.

Finally, the embeddings from both the text and graph encoders are concatenated and passed to a decoder. The decoder uses this combined representation to generate a coherent and contextually grounded natural language response. This dual-encoder approach enhances the model’s understanding by grounding its generation process in the explicit relationships of the knowledge graph.

2.3 Training and Evaluation Protocol

The model is trained using a supervised fine-tuning protocol, the workflow for which is detailed in Figure 2. The process begins with initializing the environment and loading the WebNLGGraph dataset. A GraphLLM model is initialized, with a checkpointing system in place to resume training if interrupted.

The training loop iterates over batches of graph-text pairs from the training set, with performance periodically checked against a validation set. Early stopping is implemented to monitor validation loss and prevent overfitting by halting the process if performance ceases to improve. Upon completion of training, the model checkpoint with the best validation performance is saved for final evaluation.

Model performance is assessed on the held-out test set using a suite of standard automated metrics for text generation:

- BLEU (Bilingual Evaluation Understudy) (Papineni et al., 2002) : Measures n-gram precision between the generated and reference texts, correlating with fluency and adequacy.
- METEOR (Metric for Evaluation of Translation with Explicit ORdering) (Banerjee and Lavie, 2005) : An extension of BLEU that considers synonymy and stemming, offering a more robust measure of semantic similarity.
- BERTScore (Zhang et al., 2020) : Computes the cosine similarity between token embeddings of the generated and reference texts us-

ing BERT, which captures semantic and contextual equivalence more effectively than n-gram-based metrics.

3 Performance and Results

The performance of the various model variations was comprehensively evaluated to determine the most effective architecture for the graph-to-text generation task. This evaluation is twofold: first, we analyze the validation dynamics to understand model learning behavior, and second, we assess the final output quality using quantitative metrics. The results reveal a clear performance hierarchy, where the scale of the base LLM emerged as a more critical factor than the specialized graph-aware components under these experimental conditions.

3.1 Training and Validation Dynamics

To examine each model configuration, we monitored the validation loss over 10 epochs, as shown by the curves in Figure 3.

In fact, all models successfully learned from the data, demonstrated by the consistent decrease in validation loss over the epochs. However, a significant performance gap is immediately apparent. The LLAMA 2-7b model (labeled as gt_7b in the plots) began with and maintained a substantially lower loss than all four GNN-enhanced variants. The initial validation loss for LLAMA 2-7b was already below 0.65, a level that the best-performing graph model (gt_lora) only approached by the end of its training run.

This wide margin in loss suggests that the larger, general-purpose LLM possessed a much stronger pre-trained foundation for this task. Its inherent language modeling and reasoning capabilities allowed it to adapt to the graph-to-text format more efficiently and effectively than the smaller models augmented with GNN encoders. Furthermore, the smooth decay and lower plateau of the LLAMA 2-7b validation curve indicate superior generalization to unseen graph data. Among the GNN models, the LoRA-tuned (Hu et al., 2021) variants showed a slight edge over their fully fine-tuned counterparts, hinting that parameter-efficient tuning may offer a more stable training regime for these hybrid architectures.

3.2 Quantitative Performance Evaluation

For the final performance assessment, we evaluated the models on a held-out test set using three key

Given a query at inference time:

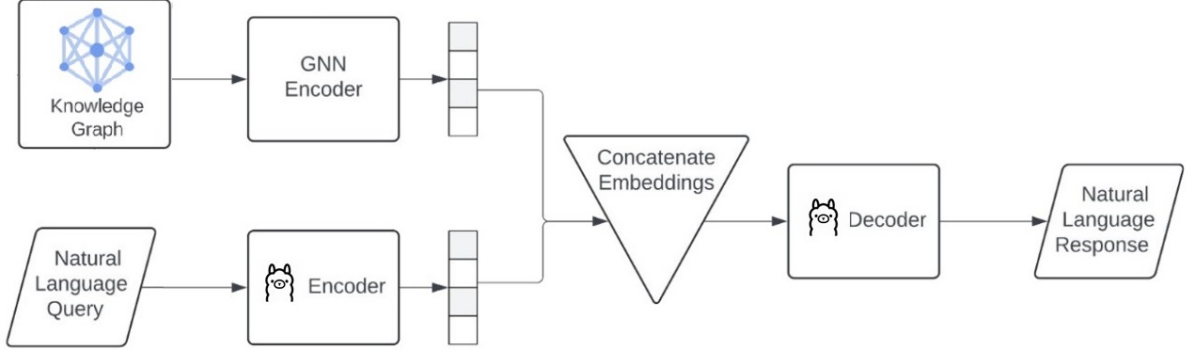


Figure 1: Architecture for knowledge-aware natural language response generation at inference time. A natural language query and a relevant subgraph from a knowledge graph are independently encoded using a language encoder and a GNN encoder, respectively. The resulting embeddings are concatenated and passed through a decoder to generate the final natural language response.

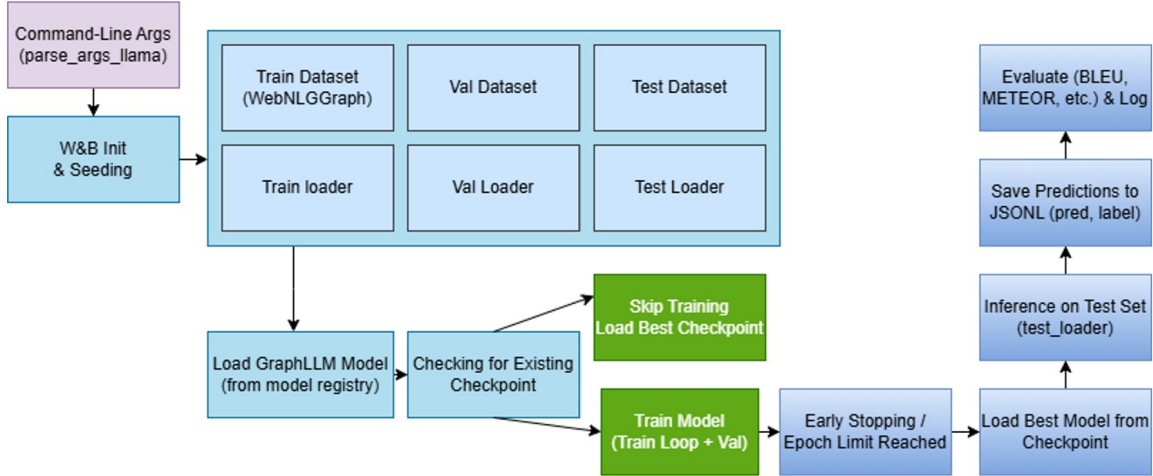


Figure 2: Training and evaluation pipeline for Graph-to-Text generation with GraphLLM. The process begins with argument parsing and W&B initialization, followed by dataset loading and model setup. Depending on checkpoint availability, the model either resumes training or loads the best checkpoint. Post-training, the best model is used for inference on the test set, with outputs saved and evaluated using standard NLG metrics.

metrics: Exact Match, BLEU, and BERTScore. The consolidated results are presented in Table 1.

The data confirms the trend observed in the loss curves: the **LLAMA2-7B + TransformerConv** model significantly outperformed all other configurations across every metric. It achieved an Exact Match score of **0.063**, a BLEU score of **49.95**, and a BERTScore of **0.958**. The high BERTScore is particularly noteworthy, as it indicates that the text generated by LLAMA2-7B + TransformerConv had the highest semantic and contextual similarity to the ground truth references. This superior perfor-

mance underscores the model’s ability to not only correctly render the facts from the graph but also to synthesize them into fluent, human-like paragraphs. For instance, the task often required synthesizing multiple disparate facts (such as details about the *Agremiação Sportiva Arapiraquense* football club, its stadium, and its nickname) into a single, coherent narrative, a challenge where the raw capability of the scaled-up base LLM proved decisive.

In contrast, the GNN-enhanced models, while competitive, did not manage to close the performance gap. The best among them, Graph -

Model	Exact Match	BLEU	BERT Score
TinyLlama + GATConv + LoRA (gat_lora)	0.053	47.69	0.954
TinyLlama + TransformerConv + LoRA (gt_lora)	0.056	48.70	0.954
TinyLlama + TransformerConv (gat)	0.049	48.48	0.952
TinyLlama + GATConv (gt)	0.052	46.08	0.952
LLAMA2-7B + TransformerConv (gt_7b)	0.063	49.95	0.958

Table 1: Performance comparison of model architectures combining LLMs (TinyLlama or LLAMA2-7B) with GNN layers (TransformerConv or GATConv). LoRA (Hu et al., 2021) indicates low-rank adaptation fine-tuning applied to TinyLlama.

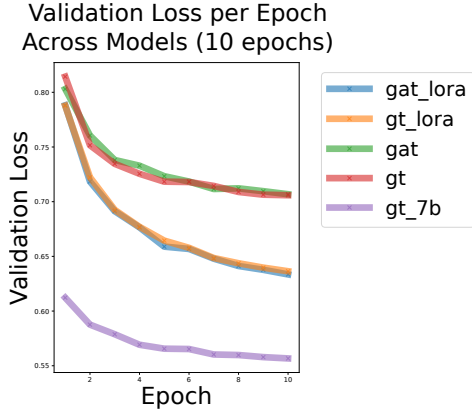


Figure 3: Validation loss curves, highlighting the superior generalization performance of the LLAMA 2-7b model.

gt_lora, achieved a BERTScore of 0.954 and a BLEU score of 48.70. Interestingly, the LoRA-tuned graph models (gat_lora and gt_lora) generally scored higher than their fully fine-tuned counterparts (gat and gt), which aligns with the observations from the training dynamics. This suggests that the structural inductive biases from the GNNs were not sufficient to overcome the sheer scale and advanced pre-training of the LLAMA 2-7b model on the WebNLG dataset. The relative simplicity of the graphs in the dataset may not have required the complex relational processing offered by a GNN, making the LLM’s powerful language generation engine the most critical component.

4 Discussion

The most striking result of this study is the superior performance of the standard fine-tuned LLAMA 2-7b model, which surpassed all GNN-enhanced configurations on every metric. This indicates that the scale and advanced pre-training of the base LLM were more critical factors for success on this task than specialized graph-aware components. The powerful language and reasoning capabilities of LLAMA 2-7b allowed it to effectively interpret the linearized graph data and generate fluent, coherent

text. The relative simplicity of the WebNLG dataset, with graphs containing only 1 to 7 triplets, likely contributed to this outcome. Such low complexity may not be sufficient to require the sophisticated relational processing that GNNs provide, making the LLM’s raw generative power the decisive component. It is also noteworthy that among the GNN models, the LoRA-tuned variants generally outperformed their fully fine-tuned counterparts, suggesting parameter-efficient tuning is a more stable approach for these hybrid architectures.

5 Conclusion

This study on graph-to-text generation using LLMs within the GEM benchmark highlights that similar model performance likely reflects evaluation limitations rather than true equivalence. Key challenges include graph linearization losses and English-centric datasets that hinder progress in low-resource languages. Advancing the field requires diverse datasets, graph-aware architectures, and refined evaluation metrics. Addressing these will unlock the broader potential of G2T systems to democratize access to structured knowledge and support global information equity.

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